MILITARY SPECIFICATION SHEET

SWITCHES, PUSH BUTTON, ILLUMINATED, 4-LAMP, LOW LEVEL TO 7.5 AMPERES, COMMON TERMINATION SYSTEM, SPLASHPROOF, SUNLIGHT READABLE, HIGH IMPACT SHOCK RESISTANT, EMI SHIELDED

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22885.

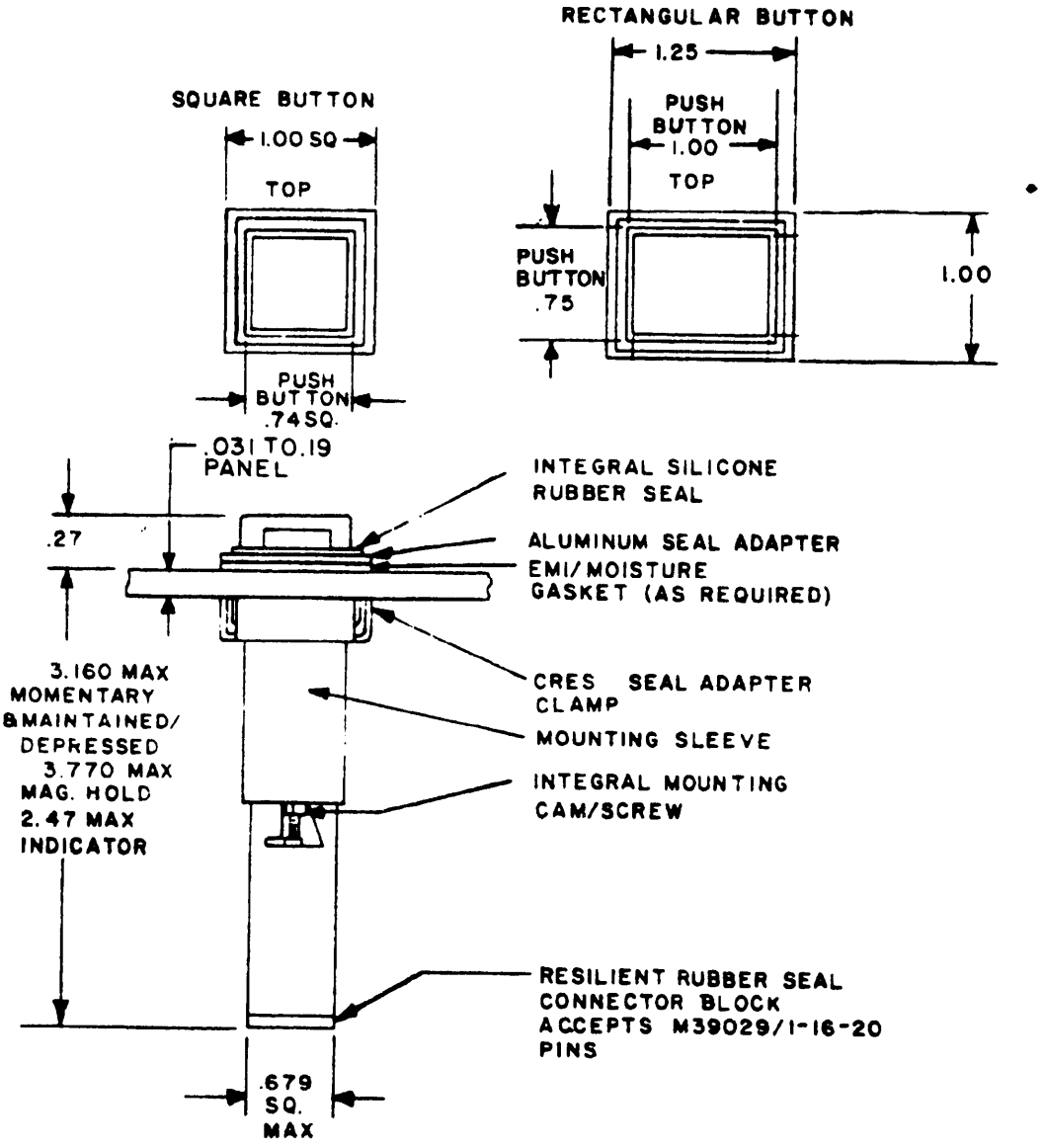
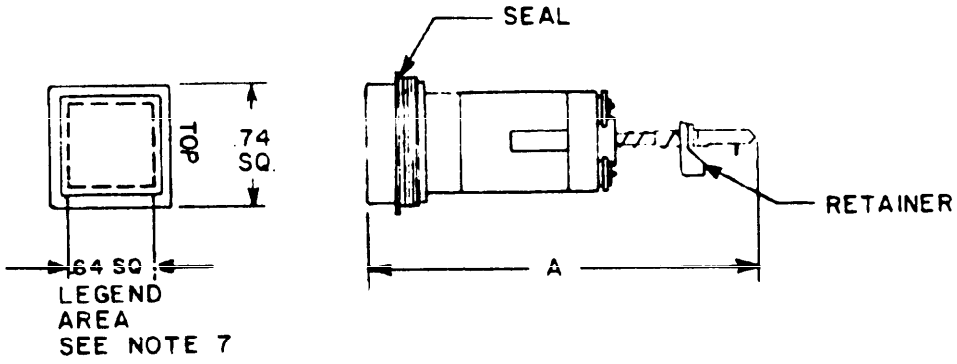


FIGURE 1. Dimensions and configurations.

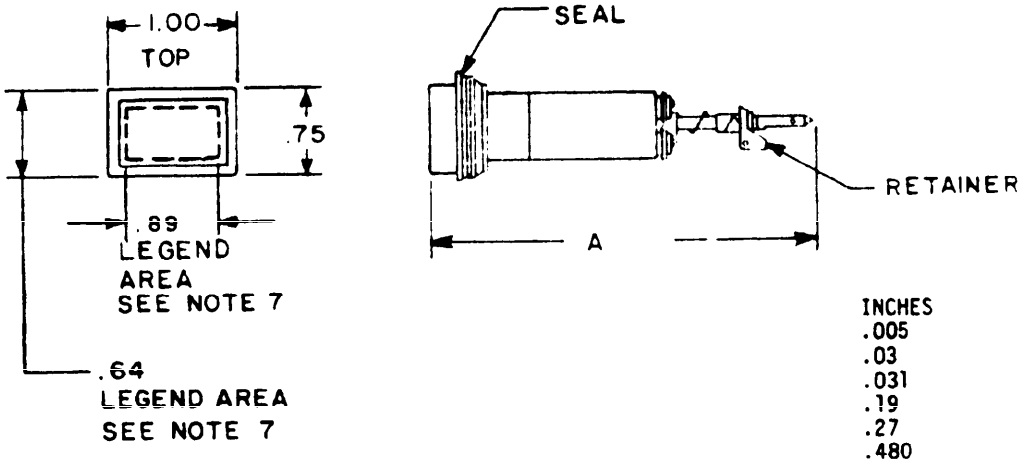
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SQUARE BUTTON



RECTANGULAR BUTTON



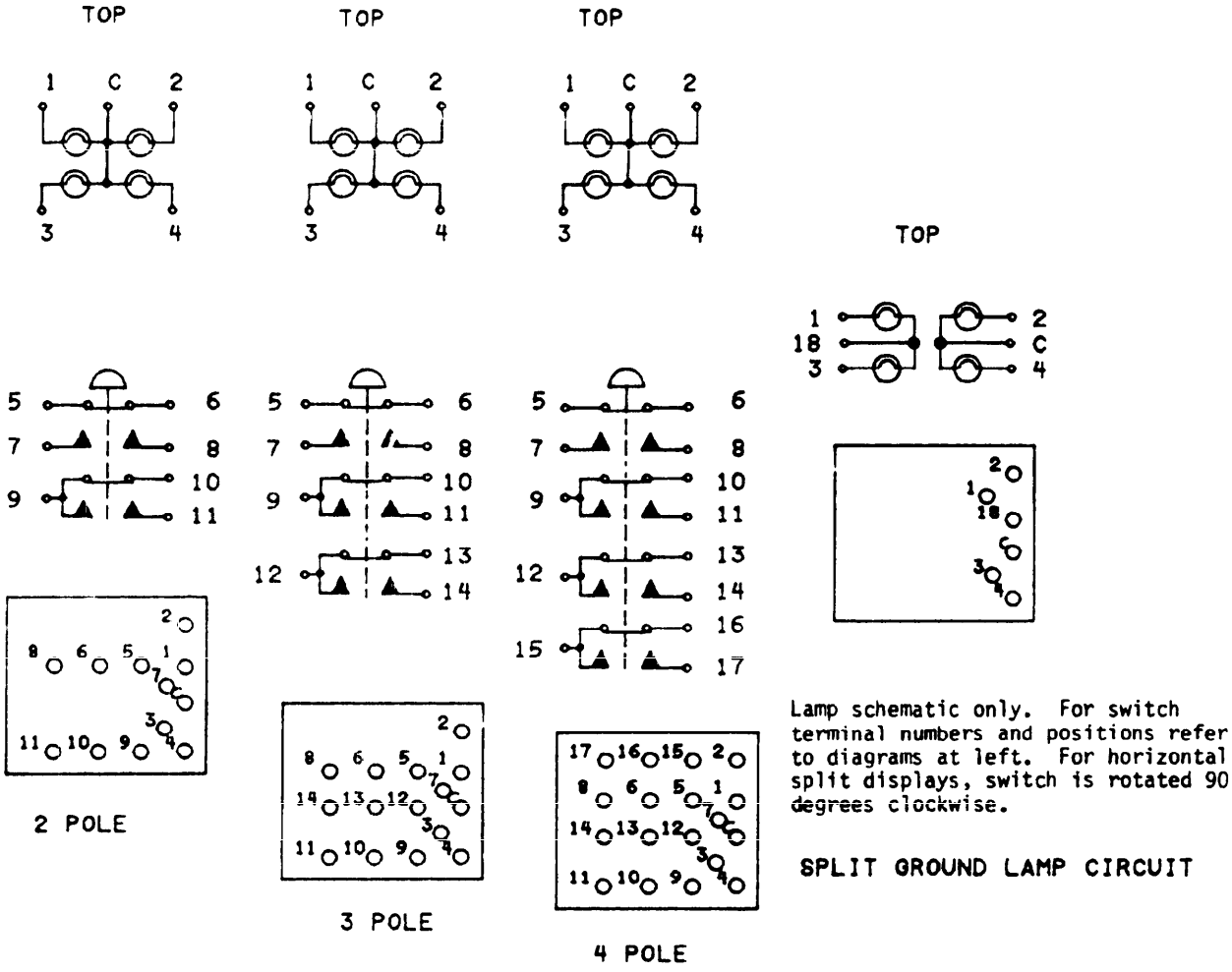
Dimension A	
Momentary	2.200 ±.030
Maintained depressed	2.200 ±.030
Indicator	2.200 ±.030
Magnetic holding	1.992 ±.030

INCHES	MM
.005	.13
.03	.76
.031	.79
.19	4.83
.27	6.86
.480	12.19
.64	16.3
.679	17.25
.74	18.8
.75	19.1
.89	22.6
1.00	25.4
1.25	31.7
3.160	80.26
3.770	95.76

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 for three place decimals and ±.03 for two place decimals.
3. Basic switches shall be MIL-S-8805/101.
4. Pushbutton shall be designed to prevent incorrect insertion into switch housing.
5. Pushbuttons shall be held captive to switch body by retaining element to prevent accidental interchange.
6. Exact shape of switch optional provided dimensions specified are not exceeded.
7. Legend area for sunlight readable display is .480 square.

FIGURE 1. Dimensions and configurations - Continued.

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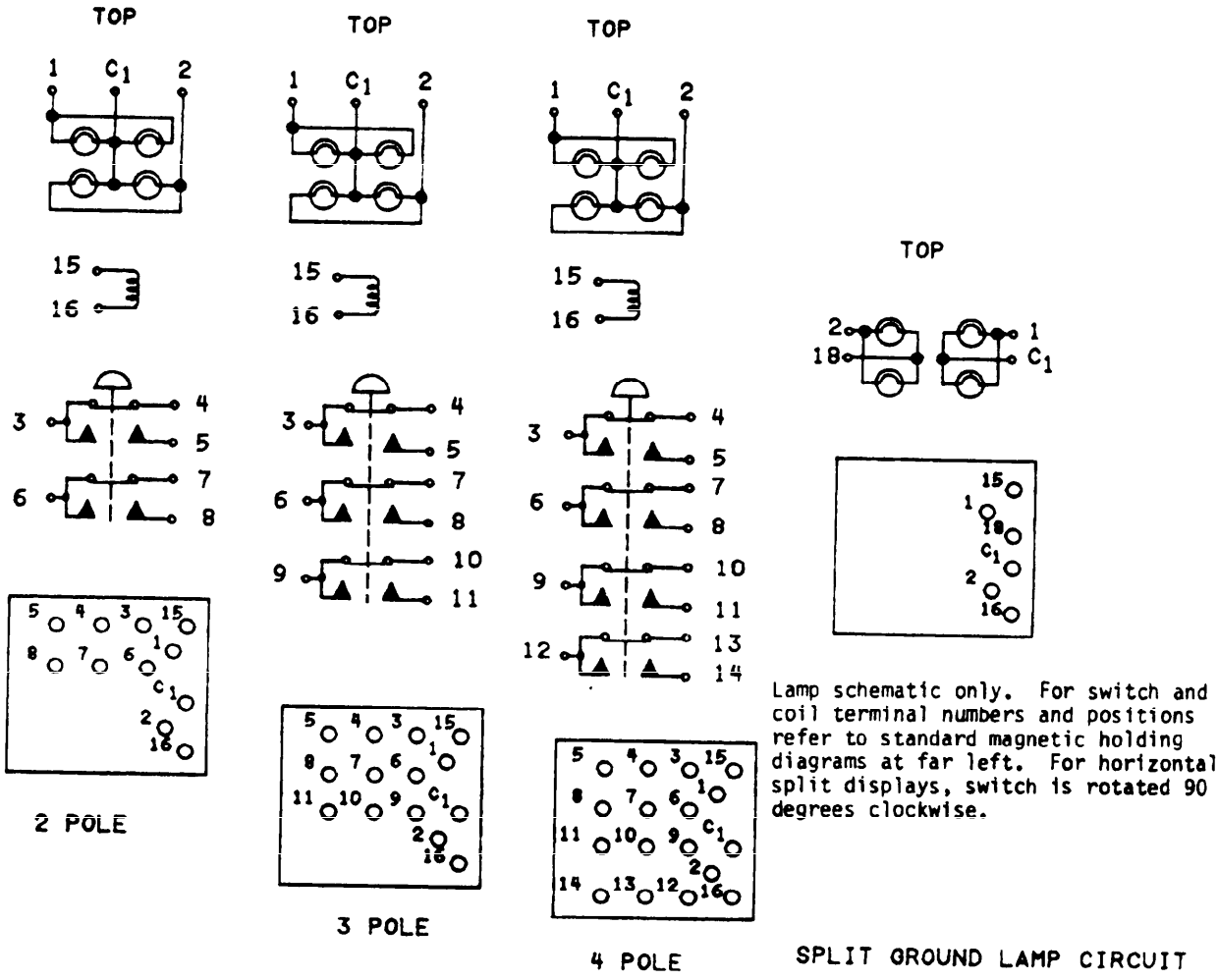


NOTE: Terminals 5 through 17 are omitted from indicator-only configurations.

View 1. Momentary maintained depressed alternate action, and indicator.

FIGURE 2. Schematic diagram.

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View 2. Magnetic holding.

NOTES:
1. Lamp schematics indicate lamp positions as viewed from front of switch.
2. Terminals shall be permanently marked on connector block as shown.

FIGURE 2. Schematic diagram - Continued.

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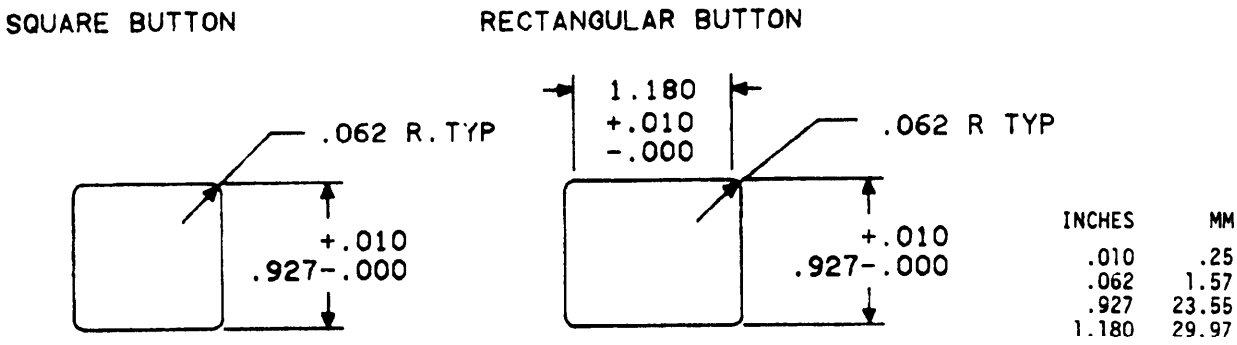


FIGURE 3. Recommended panel cutouts.

REQUIREMENTS:

Design and construction: See figures 1 thru 3.

Enclosure design: Splashproof.

Splashproof test: Test in accordance with MIL-STD-108. There shall be no leakage of water through the panel and pushbutton seals as determined by visual examination and the dielectric withstanding voltage test.

Temperature characteristic: 1 (-55° to +85°C).

Vibration grade: 3 (10-2000 Hz).

Lens transmittance: T (translucent) lens shall appear translucent white in the unlit condition. In the lit condition, it shall appear in a color determined by the color filters. Sunlight readable lens appear black in unlit condition. When lit, the legend appears in a color determined by the color filters.

Operating characteristics:

Actuation force: 3.5 ±1.5 pounds.

Actuation travel: .180 ±.015 total travel. Actuation occurs when the face of the push button is below the mounting flange.

Push button extraction force: 1.5 to 4 pounds.

Minimum hold in voltage: 18 V dc.

Maximum dropout voltage: 14 V dc.

Shock:

Method I: 75 G (MIL-STD-202, method 213, test condition H).

Method II: Applicable.

Thermal shock: During high temperature portion of thermal shock test, all four lamps shall be energized with full rated voltage.

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EMI shielding requirement: When specified, switches shall be equipped with an EMI screen. When switches are tested in accordance with the following test method, the attenuation shall not be less than specified in figure 4.

Test method: Switches shall be tested to determine shielding effectiveness in accordance with the general procedures specified in MIL-STD-285. The test shall consist of electric field and plane wave measurements performed at the following frequencies:

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
15 kHz	60 dB	100 MHz	47 dB
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB

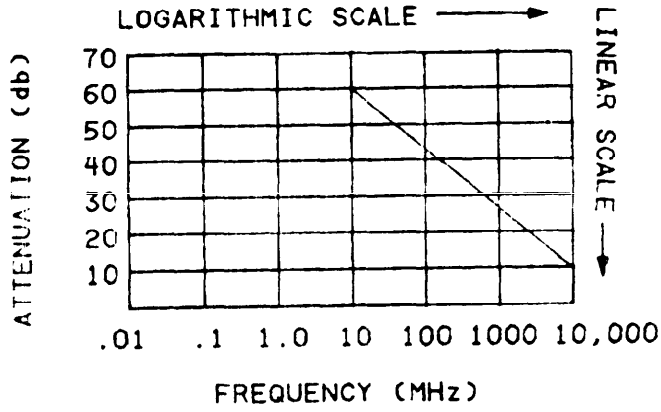


FIGURE 4. Minimum attenuation.

Weight: 2.5 ounces, maximum.

Electrical ratings:

Holding coil: .04 ±.01 amperes at 28 v dc.

Contacts: See table I.

Low level life: Applicable for gold contact switches, 50,000 cycles.

Luminance:

For other than sunlight readables: See table II.

For sunlight readable: 300 ±150 foot-lamberts (with or without EMI screen) when illuminated by 2 T-1 lamps of .15 ±.05 MSCP per half for divided legends, or 4 T-1 lamps of .15 ±.05 MSCP for full face legends. The same conditions apply for T-1 3/4 lamps of .30 MSCP minimum.

Color: In accordance with MIL-S-22885/78. See table III for sunlight readable displays.

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Terminations: Electrical connections shall be made by crimp, insertable-removable contacts and be environmentally sealed by a resilient grommet. Terminals shall meet the resistance to probe damage, maintenance aging, contact insertion and removal forces in accordance with the contact design, contact retention, and module design paragraphs of MIL-T-81714. Terminals shall accept male pin contact M39029/1-16-20.

Insertion-extraction tool: M83723/31-20 (NAS 1664-20).

Sealing grommet: Silicone rubber.

TABLE I. Electrical ratings. 1/

Contact finish	Load	Sea level		50,000 ft.
		28 V dc	115 V ac, 60 Hz	28 V dc
		(amps, max)	(amps, max)	(amps, max)
Silver	Res.	7.5	5.0	4
	Ind.	5.0	3.0	2.0
	Lamp 2/	1.0	1.0	1.0
Gold	Res.	1.0	---	1.0
	Ind.	0.5	---	0.5

1/ Minimum conditions are dependent on use and atmospheric conditions.
2/ Inrush, 10:1 for 20 milliseconds maximum.

TABLE II. Luminance (other than sunlight readable).

Color	Color symbol	Minimum average luminance 1/	
		Without EMI screen (foot lamberts)	With EMI screen 2/ (foot lamberts)
Blue	B	8.0	4.0
Green	G	50.0	25.0
Red	R	30.0	15.0
White	W	180.0	90.0
Yellow	Y	300.0	150.0

1/ When illuminated by two MS25237 lamps of .30 ±.05 mean spherical candlepower at 28 volts dc. Readings are for full display (see table VI).
2/ For information only.

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TABLE III. Illuminated color limits - sunlight readable displays. 1/

Color	Chromaticity limits	
	X	Y
Red	.695	.285
	.710	SL 2/
	.655	.325
	.660	SL 2/
Green	.300	SL 2/
	.300	.600
	.380	.600
	.360	SL 2/
Yellow	.562	.415
	.570	SL 2/
	.596	.382
	.605	SL 2/
Blue	.230	.420
	.230	.350
	.320	.350
	.320	.420
White	.400	.420
	.460	.420
	.400	.380
	.460	.380

1/ The chromaticities of the colors are expressed as "X" and "Y" coordinates on the CIE chromaticity diagram and are within the limits bound by the coordinates listed for each color.

2/ SL-spectrum locus (where intersected by other coordinate pair).

Sunlight readability: Applicable (with or without EMI shielding): This switch assembly may be obtained with legends that are visible when illuminated in direct sunlight. The legends are hidden when not illuminated. Illumination is by 4, 5 volts T-1, .15 MSCP lamps. Other T-1 lamps are available and interchangeable with different illumination results. Available area for legend illumination in all push button styles is .48 x .48 inches. The illuminated chromaticity limits are as specified in table III.

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Complete switch shall consist of:

1. Switch body - includes seal adapter, clamp, gasket and mounting sleeve.
2. Push button with integral seal.
3. Push button accepts 4 T-1 3/4 midget flange base lamps (not included, order separately). Push button with full lens style (see table VI) will accommodate lamps in the upper left and lower right positions only. Sunlight readable push button accepts 4, T-1 midget flange base lamps, or when specified, 4 T-1 3/4 midget flange base lamps.

Part numbers: Part numbers are assigned as follows:

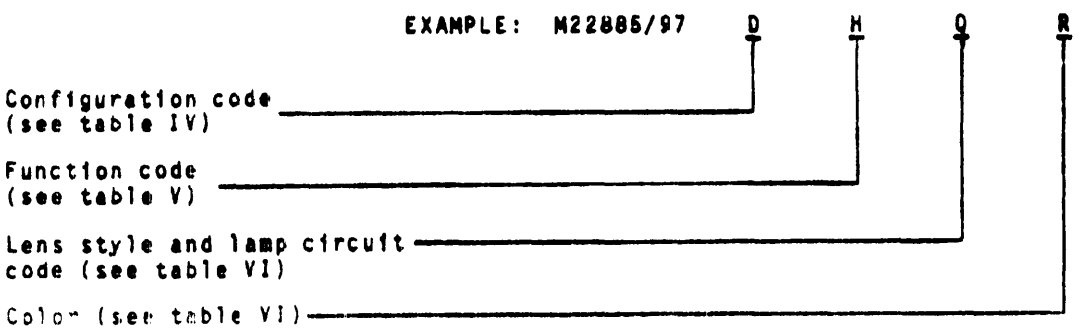


TABLE IV. Configuration code.

Configuration code	Button style (see figure 1)	EMI shielding
A	Square	Yes
B	Rectangular	Yes
C	Square	No
D	Rectangular	No

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TABLE V. Function code.

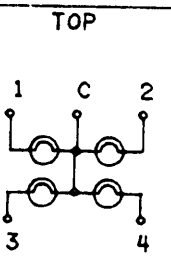
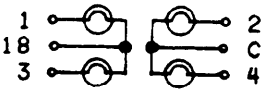
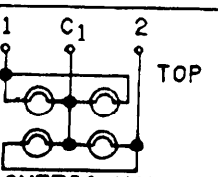
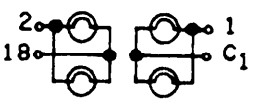
Function code	Circuit	Action	Contact material
A	2PDT	Momentary	Silver
B	3PDT		
C	4PDT		
D	2PDT	Alternate 1/	
E	3PDT		
F	4PDT		
G	2PDT	Magnet hold	
H	3PDT		
J	4PDT		
K	2PDT	Momentary	Gold
L	3PDT		
M	4PDT		
N	2PDT	Alternate 1/	
P	3PDT		
Q	4PDT		
R	2PDT	Magnet hold	
S	3PDT		
T	4PDT		
U	Indicator	-	-
V	None 2/	Momentary, Alternate, or Indicator	-
W	None 2/	Magnet hold	-

1/ Alternate action is maintained depressed.

2/ Use code V or W to specify pushbutton portion of unit only.

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TABLE VI. Lens style, lamp circuit and color.

CODE 1/	LENS STYLE 2/	LAMP CIRCUIT AND APPLICATION
A		 TOP 1 C 2 3 4 MOMENTARY AND ALTERNATE ACTION, AND INDICATOR ONLY
C		
D		
E		
F		
H		
J	NONE-SWITCH BODY ONLY	
K		 TOP 1 18 3 2 C 4 MOMENTARY AND ALTERNATE ACTION, AND INDICATOR ONLY
L	 3/	
M		
N		
P	NONE-SWITCH BODY ONLY	
Q		 1 C_1 2 TOP MAGNETIC HOLD
S	 4/	
T	NONE-SWITCH BODY ONLY	
U	 3/	 TOP 2 18 1 C_1 MAGNETIC HOLD
Z	NONE-SWITCH BODY ONLY	

1/ Codes are shown for units without color filters. When color filters are to be included, add the color symbols of table II in the sequence shown (code letters J, P, T, and Z do not require additional color filter symbols).
2/ Numbers indicate color symbol sequence.
3/ For horizontal split display, switch is rotated 90° clockwise.
4/ For vertical split display, switch is rotated 90° clockwise.

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Qualification inspection:

Group submission: When similar switches have been qualified to MIL-S-22885/77 and the basic sensitive switches are listed on QPL-8805, table VII applies.

TABLE VII. Qualification inspection (group submission).

Part number	Inspection table V of MIL-S-22885		Additional testing	Extent of approval
	Group	Number of samples		
M22885/97-DJQW	I	All		All
	II 1/2/3/	4 (from group I)		
	VII 3/	2 (from group I)	-2 samples- EMI shielding (after group VII operating characteristics tests).	
	VIII	10 4/		
Sunlight readable version of M22885/97-BJQ 5/	VIII	10 4/		
M22885/97-AFAW	- - -	2	Visual and mechanical examination operating characteristics	

1/ Insulation resistance: Following step 6 of the final cycle of the moisture resistance test, the insulation resistance measurement shall be not less than 5 megohms.

2/ Method II high impact shock only.

3/ Splashproof seal test to be performed. Twenty-five cycles of removal and reinstallation of the sealed push button shall be made as part of the intermediate current tests.

4/ Two push buttons for each color.

5/ Test legend shall fully utilize legend area. Test legend:

REMOTE
ALT LOW

Custodians:
Navy - EC

Agent:
DLA - ES

Preparing activity:
Navy - EC

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